STACKS, DIFFERENTIAL GRADED SHEAVES: LEMMA-BETTER-SET-OF-MONOS

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 4061-4083 of sources/stacks-sdga/pinned-original-source.tex; proof span is lines 4085-4181, the complete author proof (95 body lines).

Editorial note. Navigation stubs: 8 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 1 of queue_sdga.json (pb=4158, stmt 4061-4083, proof 4085-4181).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 09QK: sources/stacks-sdga/pinned-original-source.tex, source lines 4181–4181. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0AAT: sources/stacks-sdga/pinned-original-source.tex, source lines 4181–4181. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 079L: sources/stacks-sdga/pinned-original-source.tex, source lines 4181–4181. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0FSS: sources/stacks-sdga/pinned-original-source.tex, source lines 3795–3815. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0FSW: sources/stacks-sdga/pinned-original-source.tex, source lines 3995–4010. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0FST: sources/stacks-sdga/pinned-original-source.tex, source lines 3852–3860. Cited by the retained statement or proof but not part of

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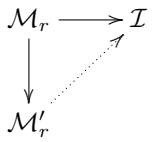
the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0FSQ: sources/stacks-sdga/pinned-original-source.tex, source lines 3731–3773. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0FS3: sources/stacks-sdga/pinned-original-source.tex, source lines 2561–2579. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $(\mathcal{C}, \mathcal{O})$ be a ringed site. Let $(\mathcal{A}, d)$ be a sheaf of differential graded algebras on $(\mathcal{C}, \mathcal{O})$. There exists a set R and for each $r \in R$ an injective map $\mathcal{M}_r \to \mathcal{M}'_r$ of acyclic differential graded $\mathcal{A}$-modules such that for an object $\mathcal{I}$ of $Mod(\mathcal{A}, d)$ the following are equivalent*

(1) *$\mathcal{I}$ is K-injective and graded injective, and*
(2) *for every solid diagram*



a dotted arrow exists in $Mod(\mathcal{A}, d)$ making the diagram commute.

Proof. Let T and $\mathcal{M}_t \to \mathcal{M}'_t$ be as in Lemma 0FSS. Let S and $\mathcal{M}_s$ be as in Lemma 0FST. Choose an injective map $\mathcal{M}_s \to \mathcal{M}'_s$ of acyclic differential graded $\mathcal{A}$-modules which is homotopic to zero. This is possible because we may take $\mathcal{M}'_s$ to be the cone on the identity; in that case it is even true that the identity on $\mathcal{M}'_s$ is homotopic to zero, see Differential Graded Algebra, Lemma 09QK which applies by the discussion in Section 0FS3. We claim that $R = T \coprod S$ with the given maps works.

The implication (1) $\Rightarrow$ (2) holds by Lemma 0FSW.

Assume (2). First, by Lemma 0FSS we see that $\mathcal{I}$ is graded injective. Next, let $\mathcal{M}$ be an acyclic differential graded $\mathcal{A}$-module. We have to show that

$$\mathrm{Hom}_{K(Mod(\mathcal{A},\mathrm{d}))}(\mathcal{M}, \mathcal{I}) = 0$$

The proof will be exactly the same as the proof of Injectives, Lemma 079L.

We are going to construct by induction on the ordinal α an acyclic differential graded submodule $\mathcal{K}_\alpha \subset \mathcal{M}$ as follows. For $\alpha = 0$ we set $\mathcal{K}_0 = 0$. For $\alpha > 0$ we proceed as follows:

(1) If $\alpha = \beta + 1$ and $\mathcal{K}_\beta = \mathcal{M}$ then we choose $\mathcal{K}_\alpha = \mathcal{K}_\beta$.
(2) If $\alpha = \beta + 1$ and $\mathcal{K}_\beta \neq \mathcal{M}$ then $\mathcal{M}/\mathcal{K}_\beta$ is a nonzero acyclic differential graded $\mathcal{A}$-module. We choose a differential graded $\mathcal{A}$ submodule $\mathcal{N}_\alpha \subset \mathcal{M}/\mathcal{K}_\beta$ isomorphic to $\mathcal{M}_s$ for some $s \in S$, see Lemma 0FST. Finally, we let $\mathcal{K}_\alpha \subset \mathcal{M}$ be the inverse image of $\mathcal{N}_\alpha$.
(3) If α is a limit ordinal we set $\mathcal{K}_\beta = \mathrm{colim}\, \mathcal{K}_\alpha$.

It is clear that $\mathcal{M} = \mathcal{K}_\alpha$ for a suitably large ordinal α. We will prove that

$$\mathrm{Hom}_{K(Mod(\mathcal{A},\mathrm{d}))}(\mathcal{K}_\alpha, \mathcal{I})$$

is zero by transfinite induction on α. It holds for $\alpha = 0$ since $\mathcal{K}_0$ is zero. Suppose it holds for β and $\alpha = \beta + 1$. In case (1) of the list above the result is clear. In case (2) there is a short exact sequence

$$0 \to \mathcal{K}_\beta \to \mathcal{K}_\alpha \to \mathcal{N}_\alpha \to 0$$

By Remark 0FSQ and since we've seen that $\mathcal{I}$ is graded injective, we obtain an exact sequence

$$\mathrm{Hom}_{K(Mod(\mathcal{A},\mathrm{d}))}(\mathcal{K}_\beta, \mathcal{I}) \to \mathrm{Hom}_{K(Mod(\mathcal{A},\mathrm{d}))}(\mathcal{K}_\alpha, \mathcal{I}) \to \mathrm{Hom}_{K(Mod(\mathcal{A},\mathrm{d}))}(\mathcal{N}_\alpha, \mathcal{I})$$

By induction the term on the left is zero. By assumption (2) the term on the right is zero: any map $\mathcal{M}_s \to \mathcal{I}$ factors through $\mathcal{M}'_s$ and hence is homotopic to zero. Thus the middle group is zero too. Finally, suppose that α is a limit ordinal. Because we also have $\mathcal{K}_\alpha = \mathrm{colim}\, \mathcal{K}_\alpha$ as graded $\mathcal{A}$-modules we see that

$$\mathrm{Hom}_{Mod^{dg}(\mathcal{A},\mathrm{d})}(\mathcal{K}_\alpha, \mathcal{I}) = \lim_{\beta<\alpha} \mathrm{Hom}_{Mod^{dg}(\mathcal{A},\mathrm{d})}(\mathcal{K}_\beta, \mathcal{I})$$

as complexes of abelian groups. The cohomology groups of these complexes compute morphisms in $K(Mod(\mathcal{A}, \mathrm{d}))$ between shifts. The transition maps in the system of complexes are surjective by Remark 0FSQ because $\mathcal{I}$ is graded injective. Moreover, for a limit ordinal $\beta \leq \alpha$ we have equality of limit and value. Thus we may apply Homology, Lemma 0AAT to conclude. □